

B. Tech. Degree VI Semester Examination April 2013**ME 602 DYNAMICS OF MACHINERY**
(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 x 5 = 40)

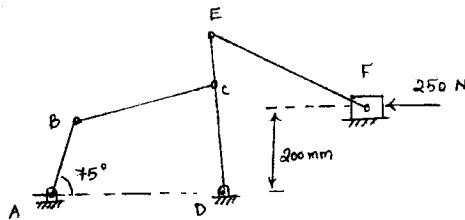
- I. (a) Discuss the effect of friction at pin joints.
 (b) Derive an expression for the velocity of a piston of a slider crank mechanism.
 (c) Explain the following terms relating to governors.
 (i) sensitiveness (ii) hunting
 (d) Find a relation for the coefficient of fluctuation of speed in terms of maximum fluctuation of energy and kinetic energy of the flywheel at mean speed.
 (e) Explain the terms : (i) variation of tractive effort (ii) swaying couple (iii) hammer blow.
 (f) Explain the method of direct and reverse cranks to determine the unbalance forces in radial engines.
 (g) With neat sketch explain the working of a single plate clutch.
 (h) With neat sketch describe the working of rope brake dynamometer.

PART B

(4 x 15 = 60)

- II. Determine the torque required to be applied to the link AB for the static equilibrium of the mechanism shown in figure. (15)

AB = 150mm, BC = 500mm, AD = 500mm, DC = 300mm, CE = 100mm and EF = 450mm.

**OR**

- III. (a) What is meant by dynamical equivalent system? Explain. (5)
 (b) The lengths of crank and connecting rod of a horizontal reciprocating engine are 200mm and 1mm respectively. The crank is rotating at 400 rpm. When the crank has turned 30° from the inner dead centre, the difference of pressure between the cover end and piston end is 0.4 N/mm^2 . If the mass of the reciprocating parts is 100kg and cylinder bore is 0.4m, calculate: (10)
 (i) inertia force
 (ii) force on piston
 (iii) piston effort
 (iv) thrust on the sides of cylinder walls
 (v) thrust in the connecting rod

Neglect the effect of piston rod diameter and frictional resistance.

- IV. The torque exerted on the crank-shaft of a two stroke engine is given by the equation $T(\text{Nm}) = 14500 + 2300 \sin 2\theta - 1900 \cos 2\theta$, where θ is the crank angle displacement from the inner dead centre. Assuming the resisting torque to be constant, determine: (15)
 (i) the power developed when speed is 150 rpm.
 (ii) the moment of inertia of the flywheel if the total fluctuation of speed is not to exceed 1% of the mean speed.
 (iii) the angular acceleration of the flywheel when the crank has turned through 30° from the inner dead centre.

OR

(P.T.O)

- V. (a) A porter governor has each of its arms of 175mm. length pivoted on the axis of the governor. The radii of rotation of the balls at the minimum and maximum speeds are 105mm and 140mm respectively. The mass of the sleeve is 20 kg and of each ball is 5 kg. Determine the range of speed when friction at the sleeve is 15N. (7)
- (b) A ship has a turbine rotor of mass 5 tonnes and radius of gyration of 60cm. It rotates at 1500 rpm in clockwise sense when looking from stern. Determine the gyroscopic couple and its effect under the following conditions. (8)
- (i) The ship steers to the left at a radius of 100m when moving at 2km/hr.
- (ii) The ship pitches with SHM with amplitude of 8 degrees and time period 12 seconds. The ship is depending with maximum velocity.
- VI. (a) What do you mean by primary and secondary unbalance in reciprocating engines? (5)
- (b) The four masses m_1, m_2, m_3 and m_4 having their radii of rotation as 200mm, 150mm, 250mm and 300mm are 200kg, 300kg, 240kg and 260kg in magnitude respectively. The angles between the successive masses are $45^\circ, 75^\circ$ and 135° respectively. Find the position and magnitude of the balance mass required, if its radius of rotation is 200mm. Compare the results using graphical and analytical method. (10)
- OR**
- VII. Determine the unbalanced forces and couples in case of following in-line engines: (15)
- (i) 2 cylinder engine
- (ii) 4 cylinder 4 stroke engine
- (iii) 6 cylinder 4 stroke engine
- VIII. (a) A band and block brake has 10 blocks and each block subtends an angle of 15° at the centre of the wheel. The two ends of the band are fixed to the pins on the opposite sides of the brake fulcrum at distances of 40mm and 200mm from it. Determine the maximum force required to be applied on the lever at a distance of 300mm from the fulcrum to absorb 250kw of power at 280rpm. The effective diameter of the drum is 840mm. Take $\mu = 0.35$. (7)
- (b) A cone clutch with a semi cone angle of 15° transmits 10kw at 600 rpm. The normal pressure intensity between the surfaces in contact is not to exceed 100 kN/m^2 . The width of the friction surfaces is half of the mean diameter. Assume $\mu = 0.25$. Determine the (8)
- (i) outer and inner diameters of the plate.
- (ii) width of the cone face and axial force to engage the clutch.
- OR**
- IX. (a) Derive an expression for the ratio of belt tensions in a flat belt drive. (7)
- (b) A leather belt transmits 8kw of power from a pulley that is 1.1m in diameter running at 200 rpm. The angle of lap is 160° and the coefficient of friction between belt and pulley is 0.25. The maximum safe working stress in the belt is 2.2 N/mm^2 . The thickness of the belt is 8mm and density of leather is 0.001 gm/mm^3 . Find the width of the belt taking centrifugal tension into account. (8)